

SWIFT SHIP TRACKER- SALESFORCE IMPLEMENTATION

SREE SASTHA INSTITUTE OF ENGINEERING AND TECHNOLOGY

Team ID : SWTID-2026-5743

Team Size : 5

Team Leader: Deliz Rishika R

Email: rdelizrishika2005@gmail.com

Team Member1: Mahalakshmi K

Email: maha904229101@gmail.com

Team Member2: Poorna Priya PS

Email: pooranapriyas@gmail.com

Team Member 3: Logesh K

Email: k.logesh1212@gmail.com

Team Member 4: Dhanush V

Email: dhanushvel2005@gmail.com

User Story:

The Swift Ship Tracker is designed to provide a centralized Salesforce-based parcel management solution that enables customers to book parcels, track delivery status, and manage shipments through an intuitive interface and conversational AI support. The system manages Parcel, Delivery, Sender, and Receiver information using Salesforce custom objects and relationships. Flows automate parcel operations and notifications, while Agentforce AI and Prompt Builder enable customers to retrieve parcel information and interact with the system conversationally. Delivery agents can update parcel status in real time, while Experience Cloud, security controls, reports, and dashboards provide secure customer access and operational visibility.

1. INTRODUCTION

1.1 Project Overview :

The Swift Ship Tracker is a Salesforce CRM solution developed to simplify parcel booking, shipment tracking, and delivery management. It supports the complete parcel lifecycle from booking and dispatch through tracking and delivery confirmation. Customers can access shipment information, while delivery agents can update parcel details and delivery status.

1.2 Purpose :

The purpose of SwiftShip Tracker is to:

- Centralize parcel and delivery information.
- Simplify parcel booking and tracking.
- Provide real-time shipment visibility.
- Automate delivery updates and customer notifications.
- Enable conversational parcel tracking using AgentForce AI and Prompt Builder.
- Improve agent efficiency through Salesforce Flows and dashboards.
- Provide secure, role-based access to customers, agents, support staff, and administrators.

2. IDEATION PHASE

2.1 Problem Statement:

PS	I am (customer)	I'm trying to	But	Because	Which makes feel
PS-1	Customer/Sender	Book a parcel easily	Booking requires support assistance or multiple application	Parcel operations are not centralized	Frustrated
PS-2	Customer / Sender	Track my parcel	I do not have a simple real-time tracking experience	shipment information is difficult to access	confused
PS-3	Delivery Agent	Update parcel status	Status updates require manual handling	there is a need for centralized operational tracking	concerned
PS-4	Customer / Receiver	Receive delivery updates	notifications may not be on available at every delivery stage	communication needs to be automated	uncertain
PS-5	Admin	Monitor parcel operations	parcel flow and delivery performance are difficult to monitor	operational data needs centralized reporting	concerned

Swift Ship Tracker Problem Statement

 Problem Statement Swift Ship Tracker – Parcel Delivery Management System 					
PS	I am	I'm trying to	But	Because	Which makes me feel
 PS-1 Customer / Sender	Customer / Sender	Book a parcel easily	Booking may require support assistance	Parcel operations are not centralized	 Frustrated
 PS-2 Customer / Sender	Customer / Sender	Track my parcel	I do not have a simple real-time tracking experience	Shipment information is difficult to access	 Confused
 PS-3 Delivery Agent	Delivery Agent	Update parcel status	Status updates may require manual handling	There is a need for centralized operational tracking	 Concerned
 PS-4 Customer / Receiver	Customer / Receiver	Receive delivery updates	Notifications may not be available at every delivery stage	Communication needs to be automated	 Uncertain
 PS-5 Admin	Admin	Monitor parcel operations	Parcel flow and delivery performance are difficult to monitor	Operational data needs centralized reporting	 Concerned

2.2 Empathy Map Canvas

User Persona: Customer / Sender

Think & Feel

- Wants a simple way to book parcels.
- Wants real-time information about shipments.
- Wants timely delivery updates.

Hear

- Delivery status notifications.
- Information from delivery agents.
- Customer support responses.

See

- Parcel ID.
- Parcel status.
- Estimated delivery date.
- Sender and receiver information.

Say & Do

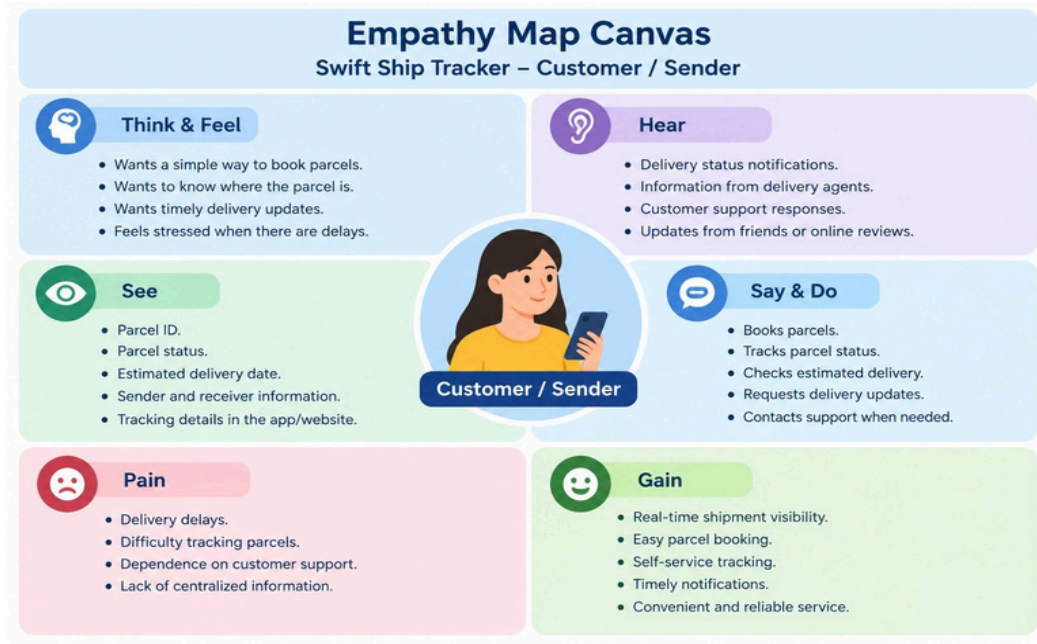
- Books parcels.
- Tracks parcel status.
- Checks estimated
- delivery. Requests
delivery updates.

Pain

- Delivery delays.
- Difficulty tracking parcels.
- Dependence on customer support.
- Lack of centralized information.

Gain

- Real-time shipment visibility.
- Easy parcel booking.
- Self-service tracking.
- Timely notifications.



2.3 Brainstorming & Idea Prioritization

Ideas Considered

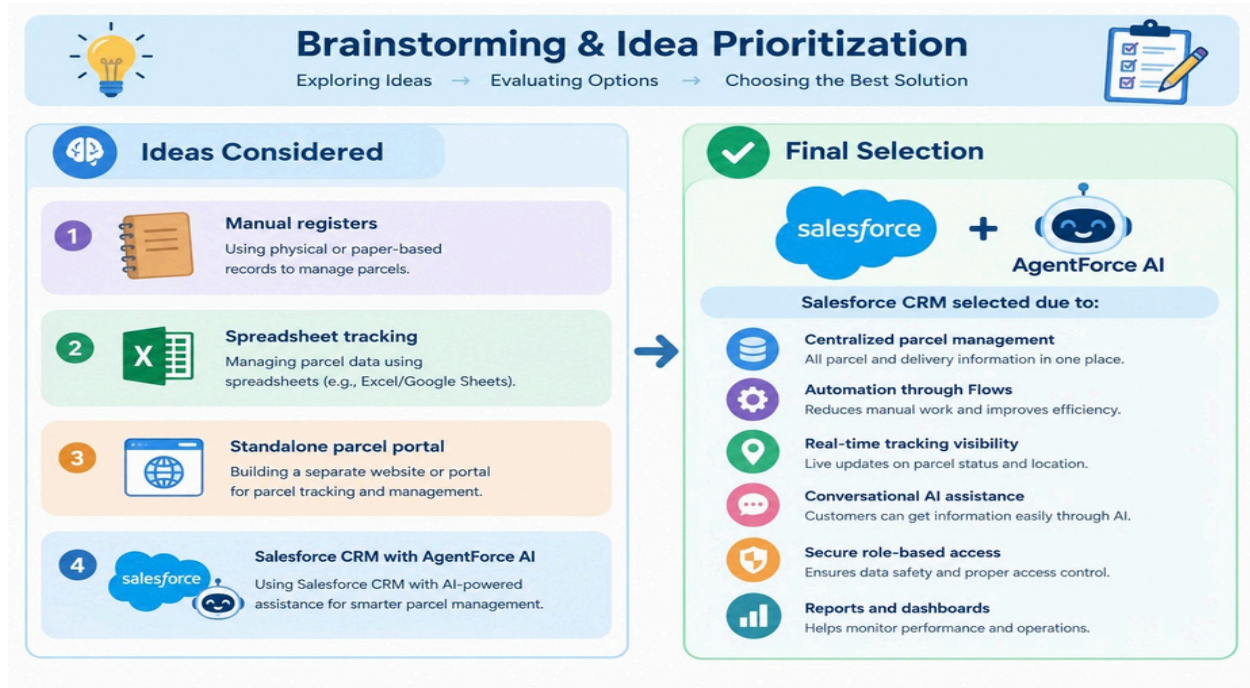
- Manual registers
- Spreadsheet tracking
- Standalone parcel portal
- Salesforce CRM with AgentForce AI

Final Selection

Salesforce CRM selected due to:

- Centralized parcel management
- Automation through Flows
- Real-time tracking visibility
- Conversational AI assistance
- Secure role-based access

- Reports and dashboards



3.REQUIREMENT ANALYSIS PHASE

3.1 Customer Journey Map

Stages:

- Parcel Booking
- Parcel Record Creation
- Dispatch & Shipment
- In-Transit Tracking
- Out for Delivery
- Delivery Confirmation
- Customer Notification
- Feedback / Support



3.2 Solution Requirements

Functional Requirements

FR No	Requirement
FR-1	Parcel booking and record creation
FR-2	Parcel status tracking
FR-3	Weight management
FR-4	Estimated delivery date management
FR-5	Sender information management
FR-6	Receiver information management
FR-7	Delivery and current-location tracking

FR-8	Agent parcel status updates
FR-9	Automated customer notifications
FR-10	AI-powered parcel tracking
FR-11	Reports and dashboards
FR-12	Customer access through Experience Cloud

Non-Functional Requirements

NFR	Description
Usability	Intuitive Salesforce and customer-facing interface
Security	Role-based and field-level access control
Reliability	Accurate parcel status and delivery information
Performance	Timely Flow and AgentForce responses
Availability	Salesforce cloud-based access
Scalability	Supports growing parcel and delivery operations
Data Integrity	Controlled access and relationship-based parcel data
Maintainability	Flows, prompts, topics, and configurations can be updated

3.3 Data Flow Diagram

Flow:

Customer / Sender



Parcel Booking



Parcel Record



Sender + Receiver Information



Delivery Record



Parcel Status Update



Current Location & Estimated Delivery



Automated Notifications



Customer Tracking / AgentForce AI



Reports & Dashboards

AI Tracking Flow

Customer Query



AgentForce AI



Prompt Builder – Retrieve Parcel Details



Parcel Record

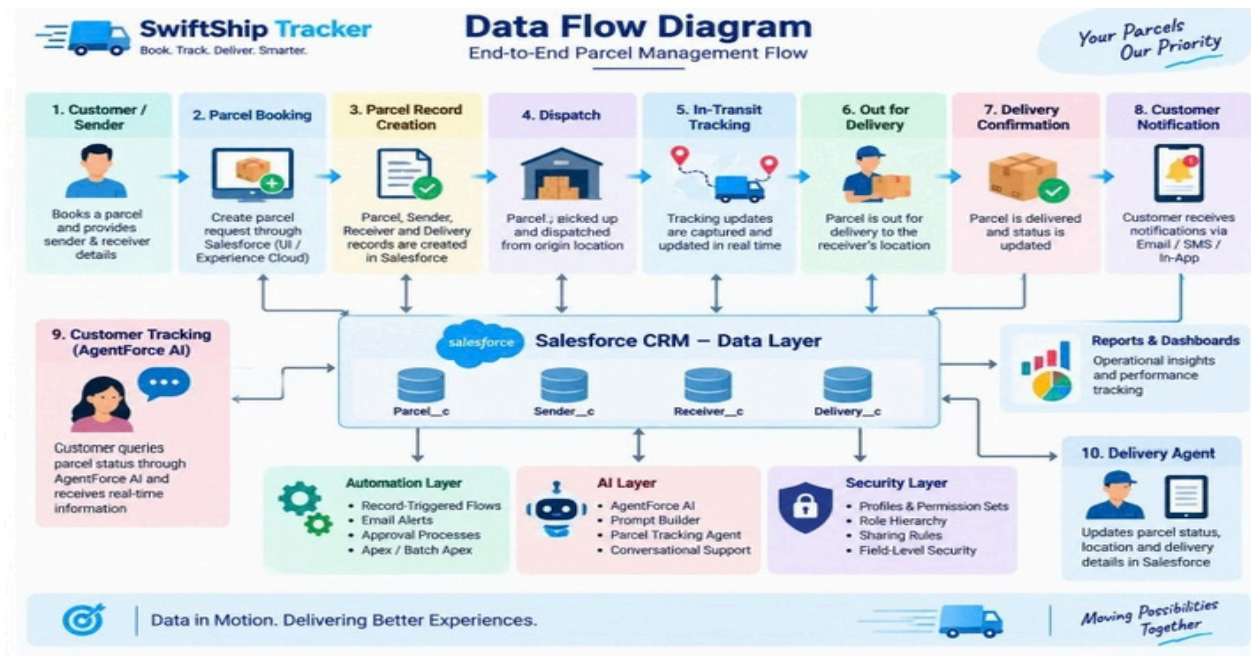


Status / Weight / Estimated Delivery



Conversational Response

The Prompt Builder implementation retrieves Parcel ID, Status, Weight, and Estimated Delivery Date information for tracking responses.



3.4 Technology Stack

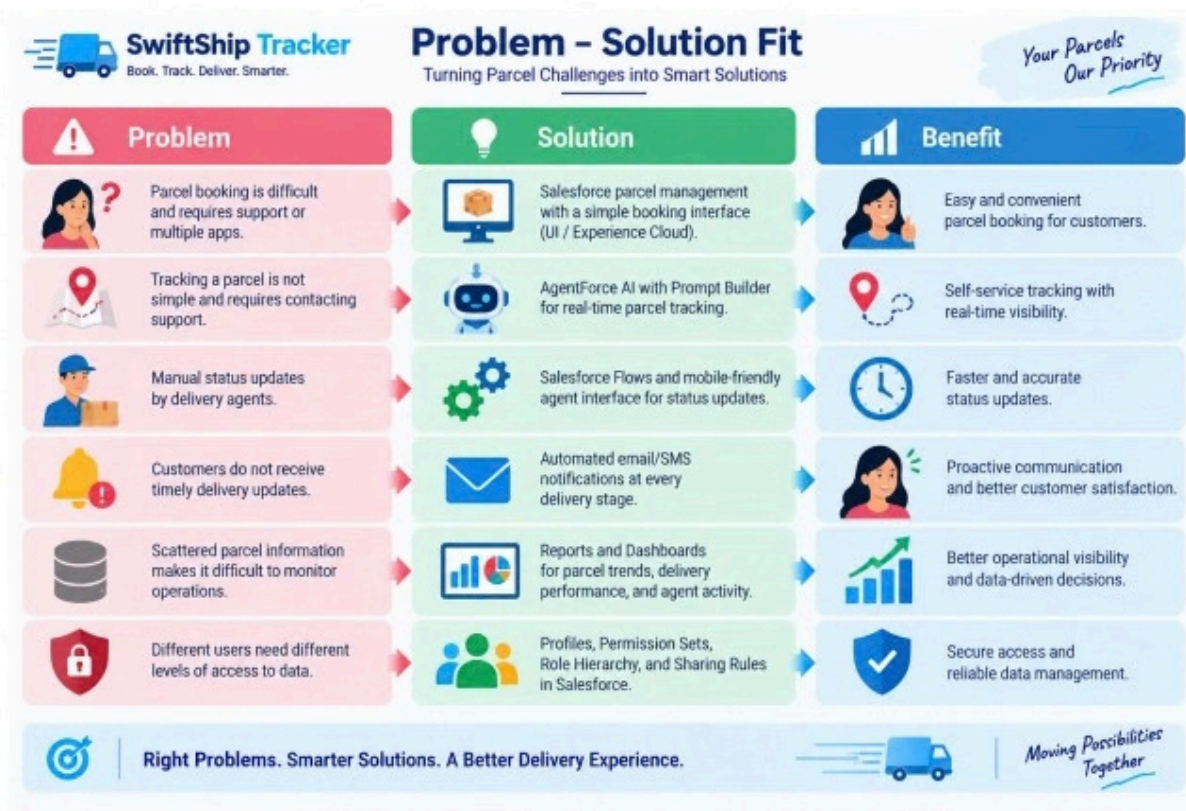
Layer	Technology
Platform	Salesforce Developer Edition
UI	Lightning App Page, Lightning Record Pages, Experience
Database	Cloud Salesforce Custom & Standard Objects
Business Logic	Salesforce Flows, Apex Classes
Automation	Record-Triggered Flows, Auto-Launched Flows, Scheduled
AI	Flows AgentForce AI, Prompt Builder
Notifications	Email Alerts & Email Templates

Processing	Batch Apex
Security	Profiles, Permission Sets, Role Hierarchy, Field-Level Security, Sharing Rules
Reporting	Reports & Dashboards

4. PROJECT DESIGN PHASE

4.1 Problem–Solution Fit

Problem	Solution	Benefit
Parcel booking is difficult	Salesforce management	Parcel Centralized booking
Tracking requires support	AgentForce AI + Prompt Self-service tracking Builder	
Manual status updates	Salesforce Flows	Automated processes
Delivery communication gaps	Email Alerts	Timely notifications
Scattered shipment information	Parcel, Delivery, Sender & Receiver objects	Structured data
Difficult operational monitoring	Reports & Dashboards	Better visibility
Different users need different access	Profiles, Roles & Permission Sets	Secure access



4.2 Proposed Solution

Functional Requirements

Paramete	Description
r Problem	Difficult and inefficient parcel booking and tracking Salesforce-based
Solution	Parcel Management System
Innovation	AgentForce AI + Prompt Builder for conversational self-service tracking and automated Salesforce Flows
Impact	Centralized booking, real-time tracking visibility, timely notifications, and efficient delivery management
Business Model	Parcel, customer, delivery-agent, support, and administrator operations
Scalability	Supports growing parcel operations, Experience Cloud, and integration with external logistics systems

4.3 Solution Architecture

- Users
- Customer | Delivery Agent | Support Staff | Admin

↓

- Salesforce Lightning / Experience Cloud

↓

- SwiftShip Tracker Salesforce Platform

- DataLayer

- Parcel
- Delivery
- Sender
- Receiver

- Automation Layer

- Flows
- Approval Processes
- Apex Classes
- Batch Apex
- Email Alerts

- AILayer

- AgentForce AI
- Prompt Builder
- Parcel Tracking Agent

- Security Layer

- Profiles
- Roles
- Permission Sets
- Sharing Rules
- Field-Level Security
- Reports & Dashboards
- Parcel Tracking & Operational Insights

5.PROJECT PLANNING & SCHEDULING

5.1Project Planning

Agile methodology

- Epics → Stories → Story Points
- Sprint-based execution

Product Backlog, Sprint Schedule, and Estimation

Sprint	Functional Requirement (Epic)	User Story Number	User Story / Task	Priority	Team Members
Sprint-1	Developer Setup	USN-1	As a system administrator, I want to create and configure the Salesforce environment so that I can build the SwiftShip Tracker parcel management system.	High	Deliz Rishika
Sprint-2	Data Modeling	USN-2	As an admin, I want to create Parcel, Delivery, Sender, and Receiver objects with proper relationships so that parcel and delivery data is stored and managed systematically.	High	Mahalakshmi
Sprint-2	Data Modeling	USN-3	As a staff member, I want dedicated tabs and a Lightning App for SwiftShip modules so that I can easily navigate and manage parcel and delivery operations	High	Poorna priya
Sprint-3	Automation	USN-4	As a system, I want to validate parcel details, mandatory fields, and booking inputs so that invalid or incomplete parcel records are prevented.	High	Logesh

Sprint-3	Automation	USN-5	As a delivery coordinator, I want automated Flows to update parcel status and delivery information so that parcel processing requires minimal manual effort.	High	Dhanush
Sprint-4	Apex	USN-6	As a system, I want Apex triggers and classes to automatically perform critical parcel and delivery actions so that important processes are handled accurately.	High	Deliz Rishika
Sprint-4	Apex	USN-7	As an inventory/operations manager, I want Batch Apex jobs to process and update parcel or delivery records automatically so that operational data remains accurate and up to date.	High	Mahalakshmi
Sprint-5	Reports	USN-8	As a manager, I want reports showing parcel status, delivery performance, and operational metrics so that I can monitor SwiftShip operations and make informed decisions.	High	Poorna priya
Sprint-6	Dashboards & AI	USN-9	As a manager, I want dashboards showing parcel status, delivery performance, and key metrics, while Agentforce provides AI-powered tracking assistance, so that I can monitor operations efficiently.	Medium	Logesh

Project Tracker, Velocity & Burndown Chart

Sprint	Total Story Points	Duration	Sprint Start Date	Sprint End Date (Planned)	Story Points Completed (as on Planned End Date)	Sprint Release Date (Actual)
Sprint-1	20	6 days				
Sprint-2	20	6 days				
Sprint-3	20	6 days				
Sprint-4	20	6 days				
Sprint-5	20	6 days				
Sprint-6	20	6 days				

6. Project Development Phase:

Phase 1: Requirement Analysis & Planning

Objective:

Customers may experience difficulty and confusion when booking, tracking, and managing parcels because they may need to contact support or use multiple applications.

SwiftShip Tracker addresses this problem by providing a centralized Salesforce platform where customers can access parcel information and use conversational AI for shipment-related queries.

Key Requirements

- Collect requirements from customers and delivery agents.
- Identify system users and their responsibilities.
- Define the system scope.
- Identify Customer, Agent, and Admin personas.
- Design relationships between Parcel, Sender, Receiver, and Delivery objects.
- Identify parcel booking and tracking use cases.
- Automate delivery-status updates.
- Use Salesforce Flows for automation.
- Use Prompt Builder for AI-based responses.
- Integrate AgentForce AI for conversational support.

DEFINING PROJECT SCOPE AND OBJECTIVES

Project Scope

1. Develop a scalable Salesforce CRM solution for parcel delivery organizations.
2. Manage customer bookings, shipments, and delivery operations.
3. Support the complete parcel lifecycle from booking to final delivery.
4. Design a structured data model for parcel-related information.
5. Implement Salesforce Flows, Apex, Email Alerts, and AI-based functionality.
6. Provide reports and dashboards for operational monitoring.
7. Provide role-based access for administrators, delivery agents, and customers.
8. Use Experience Cloud to support customer access.
9. Provide real-time parcel visibility.
10. Improve delivery-agent operations through centralized tracking.

Objectives Summary

- Automate parcel booking, tracking, and delivery notifications.
- Reduce dependency on manual updates.
- Improve customer visibility of shipment status.

- Provide self-service tracking.
- Improve delivery-agent efficiency.
- Provide centralized dashboards and reports.
- Generate useful operational insights.
- Maintain secure access using Salesforce security features.
- Improve reliability and accuracy of parcel information.

GATHERING AND ANALYSING USER NEEDS

Users Involved

Customers / Senders

Customers book parcels, track shipments, and receive delivery updates through the system or AI agent.

Delivery Agents

Delivery agents manage assigned parcels and update shipment status as the parcel moves through different delivery stages.

Customer Support Team

The support team handles delivery-related problems, responds to customer questions, and assists with communication.

Admin Team

Administrators configure Salesforce, manage user access, maintain security settings, and monitor operational performance.

System Integrator / Developer

The developer is responsible for configuring Salesforce, creating automation, developing AI functionality, and managing backend integrations.

KEY FUNCTIONAL NEEDS

The system should provide:

- Simple parcel booking and tracking.
- Customer access through Salesforce Experience Cloud.
- Agent-based parcel status updates.
- Status progression such as:
 - Booked
 - In Transit
 - Out for Delivery
 - Delivered
- Automated email notifications.
- Sender and Receiver information management.
- Contact and email information.
- AI-based parcel tracking.
- Prompt Builder integration.
- Reports and dashboards.
- Role-based access.
- Profiles and permission sets.

TOOLS USED

Salesforce Developer Org

Used for Salesforce configuration, development, and testing.

Prompt Builder / AgentForce AI

Used to create AI-assisted parcel tracking prompts and conversational responses.

Salesforce Flow Builder

Used to automate parcel booking, status changes, and delivery operations.

Google Sheets / Google Docs

Used for initial planning, requirement documentation, and system mapping.

IDENTIFYING KEY SALESFORCE FEATURES AND TOOLS

Salesforce Features

Custom Objects

- Parcel__c
- Delivery__c
- Sender__c
- Receiver__c

Standard Objects

- Account
- Contact
- Case
- Task
- EmailMessage
- User

Automation

- Record-Triggered Flows
- Auto-Launched Flows
- Scheduled Flows
- Approval Processes

Apex

- Apex Classes
- Batch Apex

Apex is used for important parcel-related automation and periodic processing.

User Interface

- Lightning App Page
- Lightning Record Pages
- Dynamic Forms
- Experience Cloud

AgentForce AI and Prompt Builder

The AI agent is configured to retrieve parcel information using the Parcel ID and provide tracking responses to customers.

Email Services

Email Alerts and Email Templates are used for:

- Booking confirmation.
- Delivery status updates.
- Delivery completion messages.

Security

The system uses:

- Profiles
- Permission Sets
- Role Hierarchy
- Field-Level Security
- Sharing Rules

These controls help provide suitable access for administrators, agents, and customers.

DESIGNING DATA MODEL AND SECURITY MODEL

Data Model

Parcel__c

The Parcel object stores the primary shipment information such as:

- Parcel ID
- Parcel Status
- Weight
- Estimated Delivery Date

It is connected to Sender and Receiver information and associated with Delivery records.

Delivery__c

The Delivery object stores information related to shipment delivery, such as:

- Current Location
- Estimated Delivery
- Parcel
- Sender

Delivery agents use this object to update and monitor delivery progress.

Sender__c

The Sender object stores:

- Sender Name
- Sender Address
- Contact Number
- Email

The sender can be associated with multiple parcels.

Receiver__c

The Receiver object stores:

- Receiver Name
- Receiver Address
- Contact Number
- Email

The receiver information is associated with the parcel and sender for delivery purposes.

IDENTIFYING KEY SALESFORCE FEATURES AND TOOLS

The SwiftShip Tracker project uses several Salesforce features and tools to develop an efficient parcel management and tracking system. Each feature is selected based on the requirements of the project.

Salesforce Custom Objects

Custom objects are created to store and manage parcel-related information. The main objects used in the system are:

- Parcel – Stores parcel details such as parcel ID, weight, status, booking information, and tracking information.
- Delivery – Stores delivery-related information including delivery status and delivery agent details.
- Sender – Stores information about the person sending the parcel.
- Receiver – Stores information about the person receiving the parcel.

These objects help organize the data and provide a centralized structure for managing parcel operations.

Lightning App and Record Pages

Lightning App Builder is used to create user-friendly pages for the SwiftShip Tracker application. Record pages display important parcel and delivery information in an organized manner.

The Lightning interface allows users to access relevant records and perform required operations efficiently.

Salesforce Flows

Salesforce Flow is used to automate business processes in the parcel management system. Flows can be used for:

- Updating parcel status.
- Automating delivery-related processes.
- Sending customer notifications.
- Updating related records.
- Reducing manual data entry.

Automation helps maintain consistency and reduces the effort required for repetitive tasks.

Experience Cloud

Experience Cloud is used to provide an external interface for customers. Customers can use the portal to access parcel-related information and track their shipments.

The portal provides a convenient way for customers to interact with the parcel tracking system without requiring direct access to the internal Salesforce application.

AgentForce AI

AgentForce AI is included to provide conversational assistance to users. Customers can interact with the AI assistant to obtain parcel-related information.

The AI assistant can help users with information such as:

- Parcel tracking status.
- Parcel weight.
- Estimated delivery information.
- General parcel-related queries.

This provides a conversational method for accessing information from the system.

Prompt Builder

Prompt Builder is used along with AgentForce to create prompts for specific parcel-related tasks. It helps define how the AI assistant should process and respond to user requests.

Reports and Dashboards

Salesforce Reports and Dashboards are used to analyze parcel and delivery information. They provide a summarized view of the data and help administrators monitor shipment activities and delivery performance.

Security Features

Salesforce security features are used to protect parcel and customer information. The project uses:

- Profiles
- Permission Sets
- Role Hierarchy
- Field-Level Security
- Sharing Rules

These features provide controlled access to records and fields based on the user's responsibilities.

Email Alerts and Templates

Email Alerts and Email Templates are used to automate customer notifications. Customers can receive updates related to parcel status and delivery activities.

This improves communication and keeps customers informed throughout the delivery process.

CONFIGURING SALESFORCE ENVIRONMENT

The Salesforce Developer Edition environment is configured as the development platform for the SwiftShip Tracker project.

The Salesforce environment is prepared by configuring the required objects, fields, relationships, user access, automation, and application components.

The main configuration activities include:

1. Creating the required custom objects.
2. Creating fields for storing parcel and delivery information.
3. Establishing relationships between objects.
4. Creating tabs for easy navigation.
5. Configuring Lightning App and Record Pages.
6. Creating Flows for process automation.
7. Configuring user permissions and security.
8. Setting up AgentForce AI and Prompt Builder.
9. Creating Reports and Dashboards.
10. Testing the complete application.

The configured environment provides the foundation for developing and testing the SwiftShip Tracker application.

CREATING CUSTOM OBJECTS AND FIELDS

Custom objects are created in Salesforce to represent the main entities of the parcel management system.

The following objects are used:

Parcel Object

The Parcel object is the central object of the system. It stores information related to individual parcels and their shipment status.

Important information includes:

- Parcel ID
- Parcel weight
- Parcel status

- Booking details
- Tracking
- information Delivery information

Delivery Object

The Delivery object manages delivery-related information. It contains information about the delivery process and delivery agent.

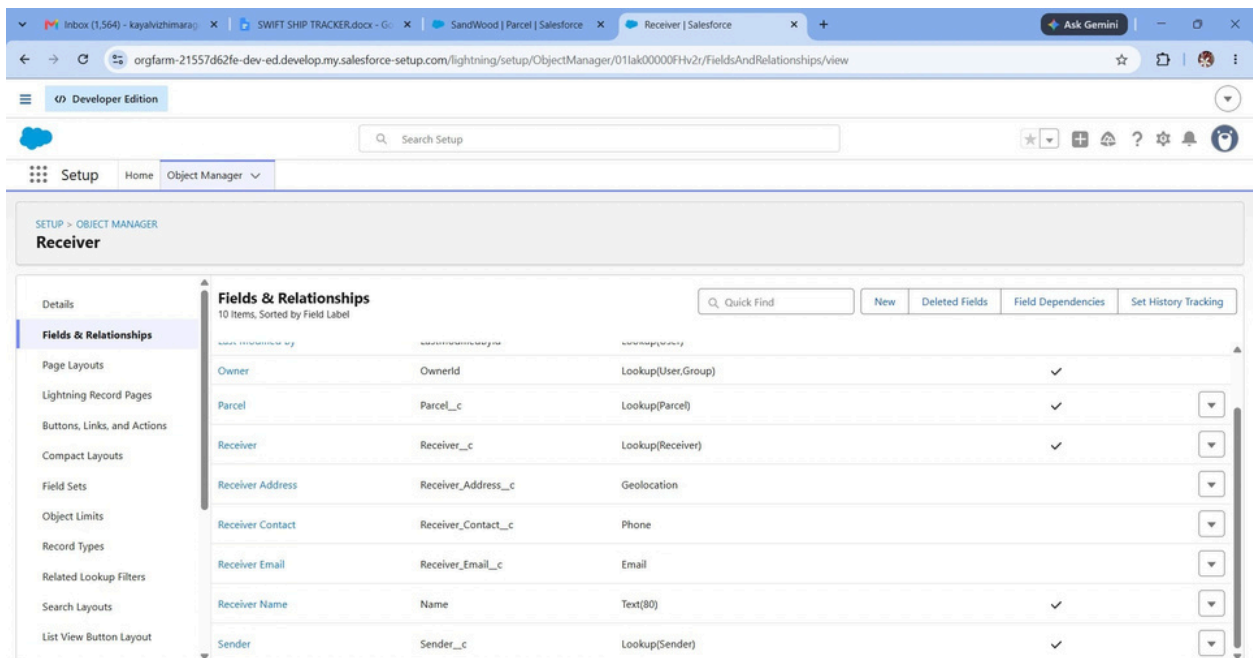
Sender Object

The Sender object stores details of the person who books or sends the parcel.

Receiver Object

The Receiver object stores details of the person who receives the parcel.

Relationships are created between the objects so that related parcel, sender, receiver, and delivery information can be accessed efficiently.



CONFIGURING AUTOMATION USING FLOWS

Salesforce Flows are configured to automate important parcel management activities.

The automation process helps reduce manual work and ensures that important actions are performed consistently.

The Flows can support activities such as:

- Parcel status updates.
- Delivery status updates.
- Customer notifications.
- Updating related records.
- Automating repetitive operations.

The use of Flows improves the efficiency of the parcel management process and helps maintain accurate information.

IMPLEMENTING SECURITY AND ACCESS CONTROL

Security is an important part of the SwiftShip Tracker system because parcel and customer information must be protected.

Salesforce security mechanisms are configured according to the responsibilities of different users.

Profiles

Profiles define the basic permissions available to users. Different users can be provided with access according to their responsibilities.

Permission Sets

Permission Sets provide additional permissions without changing the user's basic profile.

Role Hierarchy

Role Hierarchy is used to control access based on the organizational structure.

Field-Level Security

Field-Level Security controls which fields users can view or edit. Sensitive information can therefore be protected from unauthorized access.

Sharing Rules

Sharing Rules are used to provide appropriate record-level access to users when required.

The combination of these security features helps provide controlled and secure access to the SwiftShip Tracker system.

PHASE 2: SALESFORCE DEVELOPMENT – BACKEND & CONFIGURATIONS

This phase focused on developing the core logic of SwiftShip Tracker. Custom objects were created for Parcel, Sender, Receiver, and Delivery with relationships. Automations were built using Flows for parcel status updates and approval processes.

Milestone 1 – Salesforce Account

Introduction

Are you new to Salesforce? Not sure exactly what it is, or how to use it? Don't know where you should start on your learning journey? If you've answered yes to any of these questions, then you're in the right place. This module is for you.

Welcome to Salesforce! Salesforce is game-changing technology, with a host of productivity-boosting features, that will help you sell smarter and faster. As you work toward your badge for this module, we'll take you through these features and answer the question, "What is Salesforce, anyway?"

What Is Salesforce?

Salesforce is your customer success platform, designed to help you sell, service, market, analyse, and connect with your customers.

Creating Developer Account

Creating a developer org in Salesforce.

1. Go to the Salesforce Developer signup page using <https://developer.salesforce.com>
2. On the sign-up form, enter the required details:
 - First Name & Last Name
 - Email
 - Job Title: Developer
 - Company: College Name
 - Country: India
3. Click **Sign Me Up** after filling in the details.

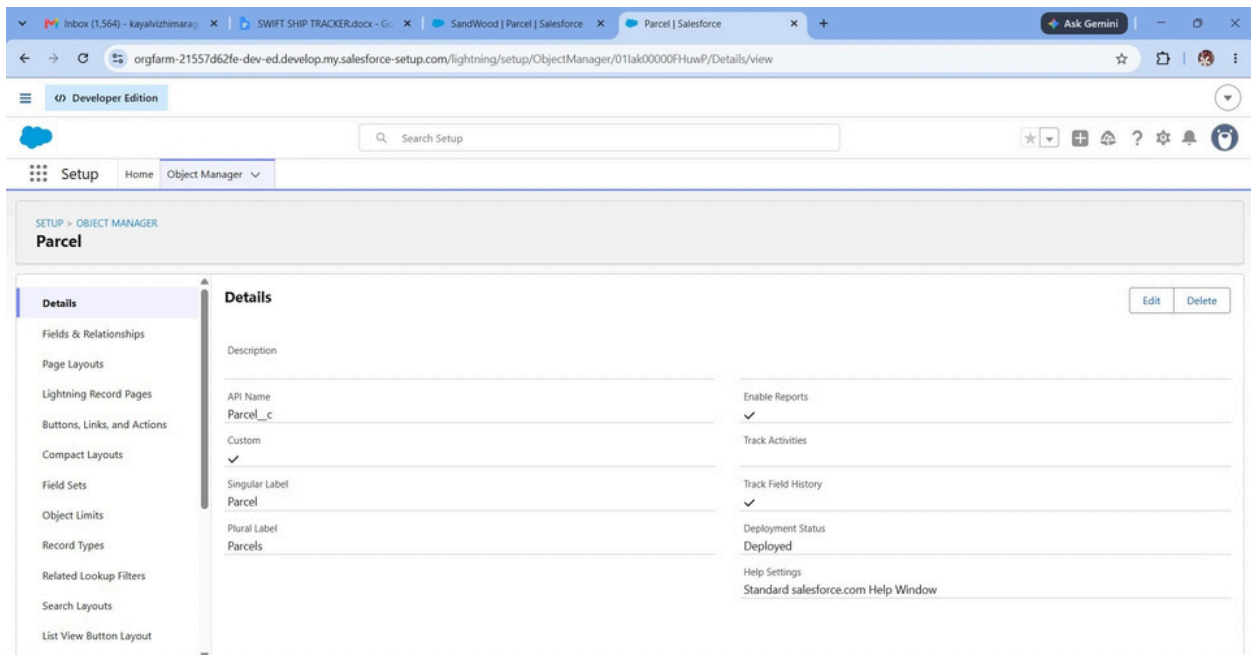
4. Go to the inbox of the email that you used while signing up.
5. Click on **Verify Account** to activate your account.
6. The email may take 10–30 minutes and sometimes up to 2 hours.
7. Give a password and answer a security question.
8. Click on **Change Password**.
9. You will then be redirected to your Salesforce Setup page.

Milestone 2 – Objects Creation

Creating Parcel Object

To create an object:

1. From the Setup page, click on **Object Manager**.
2. Click on **Create** → **Custom Object**.
3. Enter the Label Name as **Parcel**
4. Enter the Plural Label Name as **Parcels**.
5. Enter Record Name as **ParcelName**.
6. Select Data Type as **Text: ParcelName**.
7. Select **Allow Reports**
8. Select **Allow Search**
9. Select **Allow Track Field History**.
10. Click **Save and New**

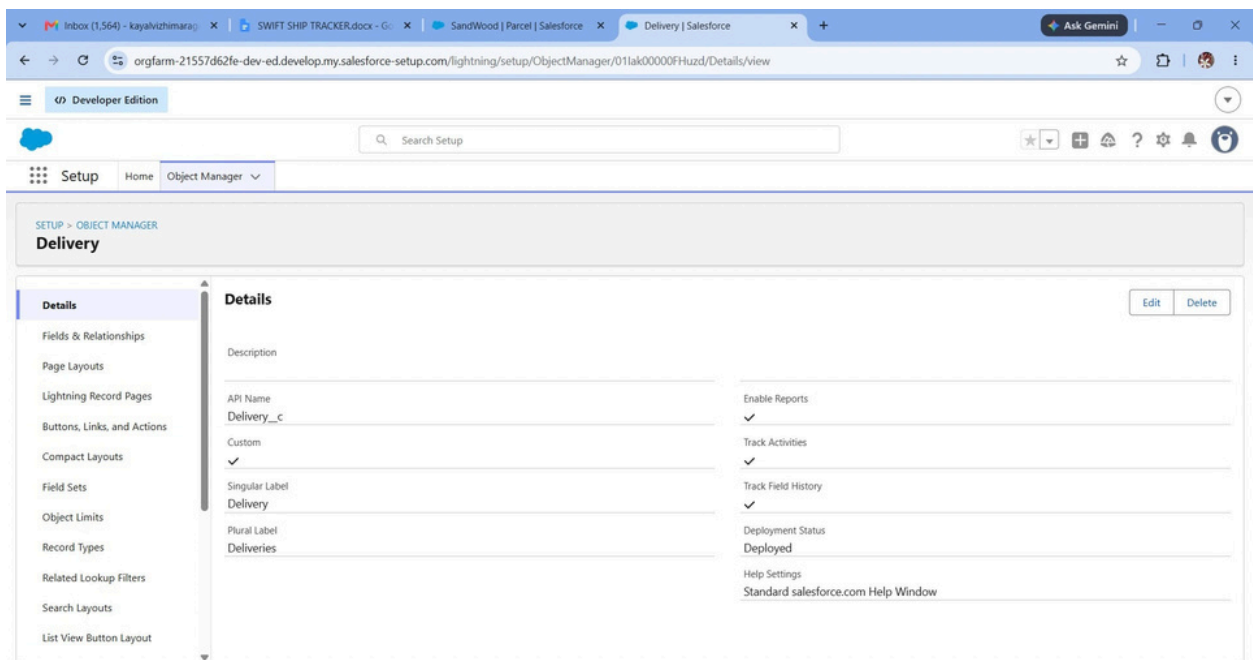


The screenshot shows the Salesforce Setup page for the 'Parcel' object. The browser address bar indicates the URL: `orgfarm-21557d62fe-dev-ed.develop.my.salesforce-setup.com/lightning/setup/ObjectManager/01ak00000FHuxwP/Details/view`. The page title is 'Developer Edition'. The navigation bar includes 'Setup', 'Home', and 'Object Manager'. The main content area is titled 'SETUP > OBJECT MANAGER' and 'Parcel'. On the left, a sidebar lists various configuration options: Details, Fields & Relationships, Page Layouts, Lightning Record Pages, Buttons, Links, and Actions, Compact Layouts, Field Sets, Object Limits, Record Types, Related Lookup Filters, Search Layouts, and List View Button Layout. The 'Details' section is active, showing a 'Description' field. The 'API Name' is 'Parcel_c'. The 'Custom' checkbox is checked. The 'Singular Label' is 'Parcel' and the 'Plural Label' is 'Parcels'. On the right, there are checkboxes for 'Enable Reports' (checked), 'Track Activities', 'Track Field History' (checked), and 'Deployment Status' (Deployed). At the bottom right, there is a 'Help Settings' link and a 'Standard salesforce.com Help Window' link. 'Edit' and 'Delete' buttons are visible in the top right corner of the details section.

Creating Delivery Object

To create an object:

1. From the Setup page, click on **Object Manager**.
2. Click on **Create** → **Custom Object**.
3. Enter the Label Name as **Delivery**.
4. Enter the Plural Label Name as **Deliveries**.
5. Enter Record Name as **DeliveryName**.
6. Select Data Type as **Text: DeliveryName**.
7. Select **Allow Reports**
8. Select **Allow Search**
9. Select **Allow Track Field History**.
10. Click **Save and New**

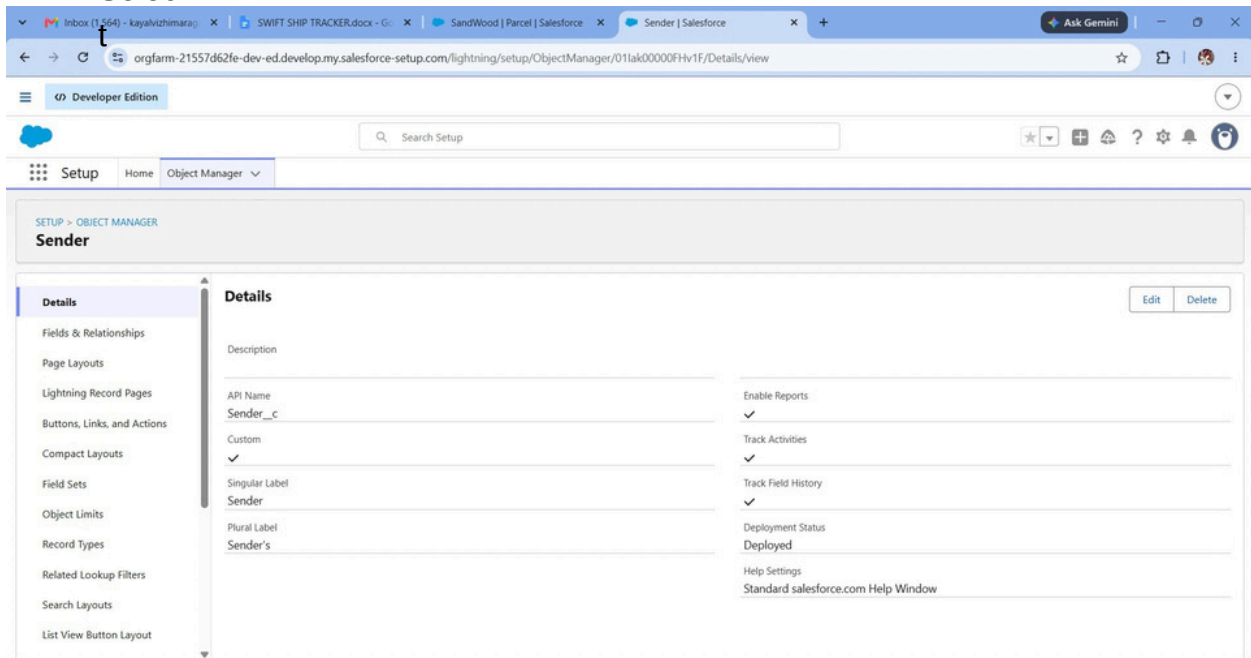


Creating Sender Object

To create an object:

1. From the Setup page, click on **Object Manager**.
2. Click on **Create** → **Custom Object**.
3. Enter the Label Name as **Sender**.
4. Enter the Plural Label Name as **Senders**.

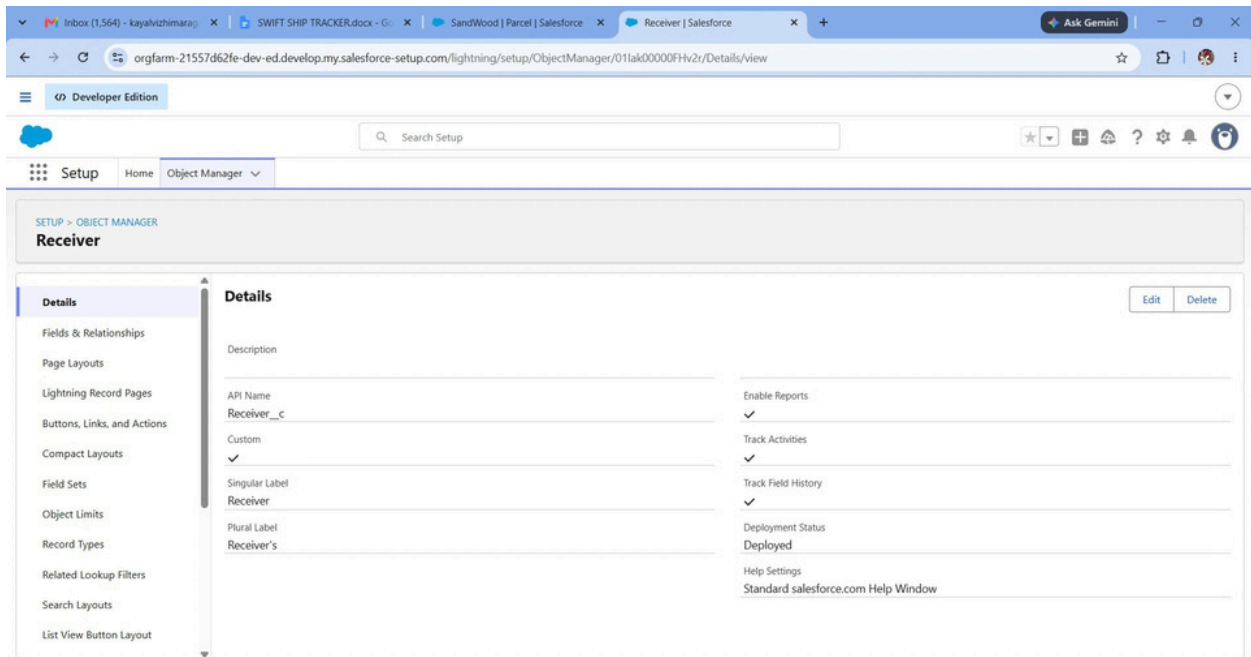
5. Enter the required Record Name.
 6. Select the appropriate data type.
 7. Select **Allow Reports**
 8. Select **Allow Search**
 9. Select **AllowTrackFieldHistory**.
 10. Click **Save and New**
- Select



Creating Receiver Object

To create an object:

1. From the Setup page, click on **Object Manager**.
2. Click on **Create** → **CustomObject**.
3. Enter the Label Name as **Receiver**.
4. Enter the Plural Label Name as **Receivers**.
5. Enter the required Record Name.
6. Select the appropriate data type.
7. Select **Allow Reports**
8. Select **Allow Search**
9. Select **AllowTrackFieldHistory**.
10. Click **Save and New**



Milestone 3 – Tabs

Creating a Tab for Parcel Object

1. Go to the **Setup** page → type **Tabs** in the Quick Find bar.
2. Click on **Tabs**.
3. Click on **New** under Custom Object Tabs.
4. Select Object (**Parcel**) → Select the Tab Style.
5. Click **Next** → under Add to Profiles, keep it as default → Click **Next**.
6. Under Add to Custom App, uncheck **Include Tab**.
7. Make sure that **Append tab to the user's existing personal customizations** is checked.
8. Click **Save**.

More Tab Creation

1. Go to **Setup** → **Tabs** and type **Tabs** in the QuickFind box.
2. Click **Tabs** under the User Interface section.
3. Click **New** under Custom Object Tabs.
4. Select the object **Delivery, Sender, or Receiver** choose a Tab Style.
5. Click **Next** → keep the Add to Profiles settings as default → Click **Next**
6. On the Add to Custom Apps page, uncheck **Include Tab**
7. Ensure **Append tab to the user's existing personal customizations** is checked.

- Click **Save** and repeat the steps for each object **Delivery, Sender, Receiver**

Milestone 4 – Fields & Relationships

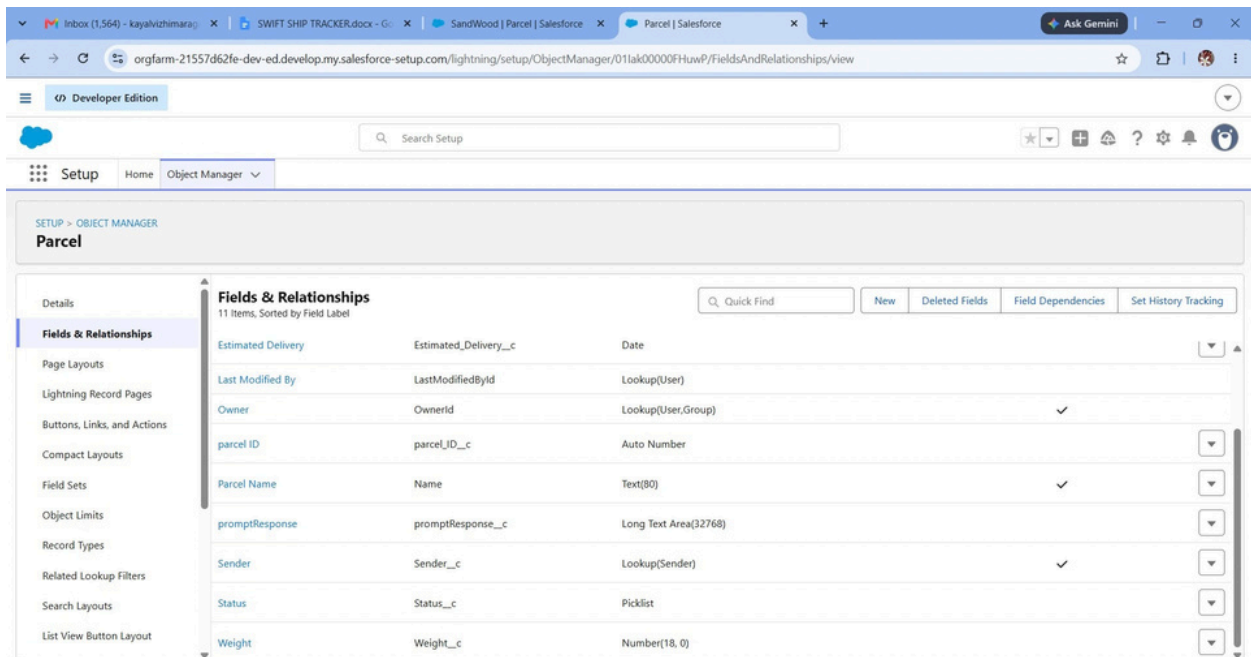
Create fields and relationships for the custom objects created in the previous milestone.

Creating Fields and Relationships for Parcel Object

- Beside the Quick Find box, click **Object Manager**.
- Search for and select your **Parcel** object.
- Go to **Fields & Relationships** → click **New**.

Field 1: Parcel ID

- Choose Data Type **Auto Number**
- Display Format **P-{000}**
- Click **Next** → **Next** → **Save & New**.
- Follow the same steps to create the remaining fields (**Status, Weight, Estimated Delivery, Sender** of the Parcel object mentioned in the table in the document.



Creating Fields and Relationships for Delivery Object

- Beside the Quick Find box, click **Object Manager**.

2. Search for and select your **Delivery** object.
3. Go to **Fields & Relationships** → click **New**.

Field 1: Current Location

- Choose Data Type: **Geolocation**
 - Click **Next**
 - Field Label: **Current Location**
 - Click **Next** → **Next** → **Save & New**.
4. Follow the same steps to create the remaining fields (**Estimated Delivery**, **Sender**, **Parcel**) of the Delivery object mentioned in the table in the document.

FIELD LABEL	FIELD NAME	DATA TYPE	CONTROLLING FIELD	INDEXED
Created By	CreatedBy	Lookup(User)		
Current Location	Current_Location__c	Geolocation		
Delivery Name	Name	Text(80)		✓
Estimated Delivery	Estimated_Delivery__c	Date		
Last Modified By	LastModifiedBy	Lookup(User)		
Owner	OwnerId	Lookup(User,Group)		✓
Parcel	Parcel__c	Lookup(Parcel)		✓
Sender	Sender__c	Lookup(Sender)		✓

Creating Fields and Relationships for Sender Object

1. Beside the Quick Find box, click **Object Manager**.
2. Search for and select your **Sender** object.
3. Go to **Fields & Relationships** → click **New**.

Field 1: Sender Address

- Choose Data Type: **Geolocation**
- Click **Next**.
- Field Label: **Sender Address**

- Click **Next** → **Next** → **Save & New**.
4. Follow the same steps to create the remaining fields **Sender Contact, Sender Email** of the Sender object mentioned in the table in the document.

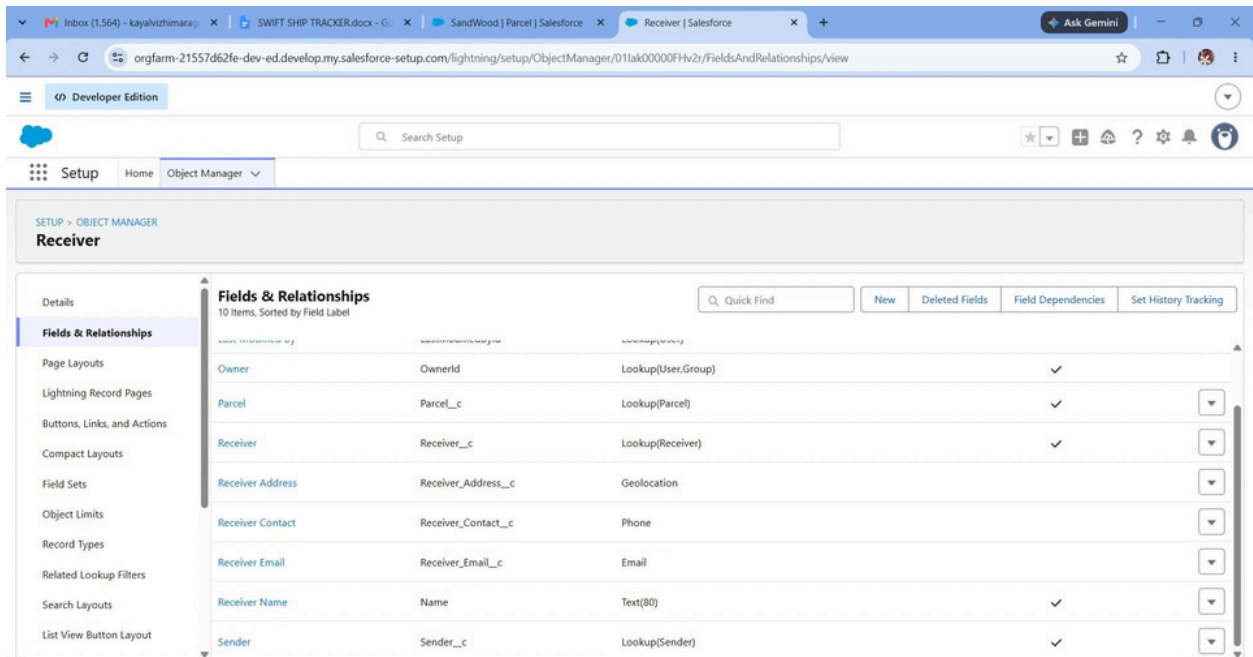
FIELD LABEL	FIELD NAME	DATA TYPE	CONTROLLING FIELD	INDEXED
Created By	CreatedById	Lookup(User)		
Last Modified By	LastModifiedById	Lookup(User)		
Owner	OwnerId	Lookup(User,Group)		✓
Sender Address	Sender_Address__c	Geolocation		
Sender Contact	Sender_Contact__c	Phone		
Sender Email	Sender_Email__c	Email		
Sender Name	Name	Text(80)		✓

Creating Fields and Relationships for Receiver Object

1. Search for and select your **Receiver** object.
2. Go to **Fields & Relationships** → click **New**.

Field 1: Receiver Address

- Choose Data Type **Geolocation**
 - Click **Next**
 - Field Label: **Receiver's Address**
 - Click **Next** → **Next** → **Save & New**.
3. Follow the same steps to create the remaining fields **Receiver Contact, Receiver Email, Sender, Parcel** of the Receiver object mentioned in the table in the document.



Milestone 5 – Prompt Builder

Designing Prompts for Agent Building

In Salesforce, go to **Setup**.

1. In the Quick Find box, type **Prompt Templates** (or in Agent Console, go to **Agent Builder** → **Prompt Templates**).
2. Click **New Prompt Template**.
3. Select **Record Summary Template**.

Enter the Details

4. Enter the following details:
 - **Label:** RetrieveParcelDetails
 - **API Name:** Retrieve_Parcel_Details
 - **Description:** Retrieves parcel details where Parcel_ID matches the user input.

Enter the Prompt Template Content

5. In the Prompt Template Text area, paste the following:

Retrieve details from Parcel c where Parcel_ID__c matches user input. Return response in this format:

Parcel Tracking Update

Parcel Tracking Update

- Parcel Name: {{\$Input:Parcel__c.Name}}
- Parcel ID: {{\$Input:Parcel__c.Parcel_ID__c}}
- Status: {{\$Input:Parcel__c.Status__c}}
- Weight: {{\$Input:Parcel__c.Weight__c}}
- Estimated Delivery Date: {{\$Input:Parcel__c.Estimated_Delivery_Date__c}}

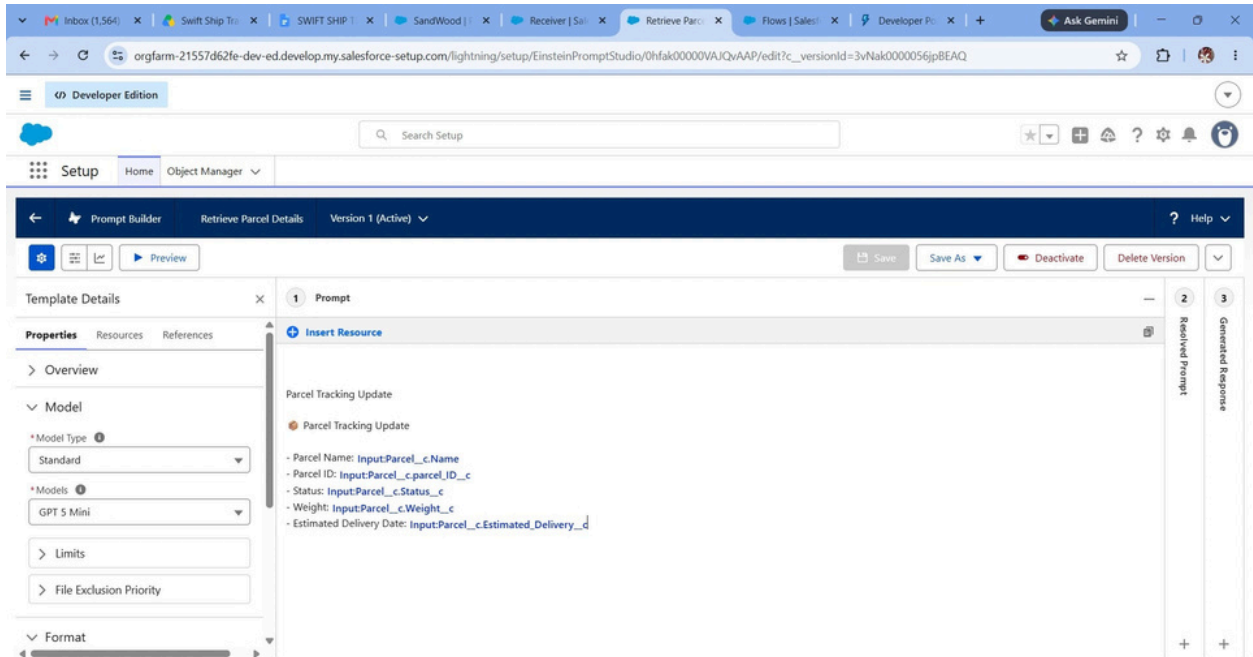
Add Resource Mapping

6. Scroll down to the **Resources / Insert Resource** section.
7. Click **Insert Resource** → **Object**
8. Select **Parcel__c**
9. Add the following fields:
 - Name
 - Parcel_ID__c
 - Status__c
 - Weight__c
 - Estimated_Delivery_Date__c
10. Click **Insert Resource** → **Add** for each one.
11. Click **Save**
12. Then click **Activate** to make this Prompt Template available for your agent or flow.

Testing the Prompt

Test the Prompt:

1. Open your Agent preview.
2. Type: **“Track parcel P-12345”**.
3. It should return details from the Parcel record.



Milestone 6 – Flows

Create flows for capturing the response as shown.

1. Goto **Setup** → **search Flows** → **click New Flow**
2. Choose **Auto-Launched Flow (No Trigger)** click **Create**.

Add Input Variables

3. Add input variables to allow the Agent to pass values:

- Click **New Resource** → **Variable**.
- Select **Available for Input**.
- **Ids**
 - Data Type: Text
 - Available for Input
- **Output**
 - Data Type: Text
 - Available for Output

Add Get Records Element

4. Click on the **Add** icon to add an element:

Get Records

- Label: **Get Parcel Records**
- Object: **Parcel**
- Field: **Parcel_ID__c**
- Operator: **Equals**
- Value: **Ids**

Add Action Element

5. Add an **Action** element.
6. Find **Retrieve Parcel Details**.
7. Click **Retrieve Parcel Details**.
8. Label: **Parcel Updates**.
9. Object to Summarize: **Parcel from Get Parcel Records**.

Add Assignment Element

6. Add an Assignment element.
- Label: Assignment
- Variable: Output
- Operator: Equals
- Value: Outputs from Action → Prompt Response

Connect the Flow

7. Connect the elements in the following order:

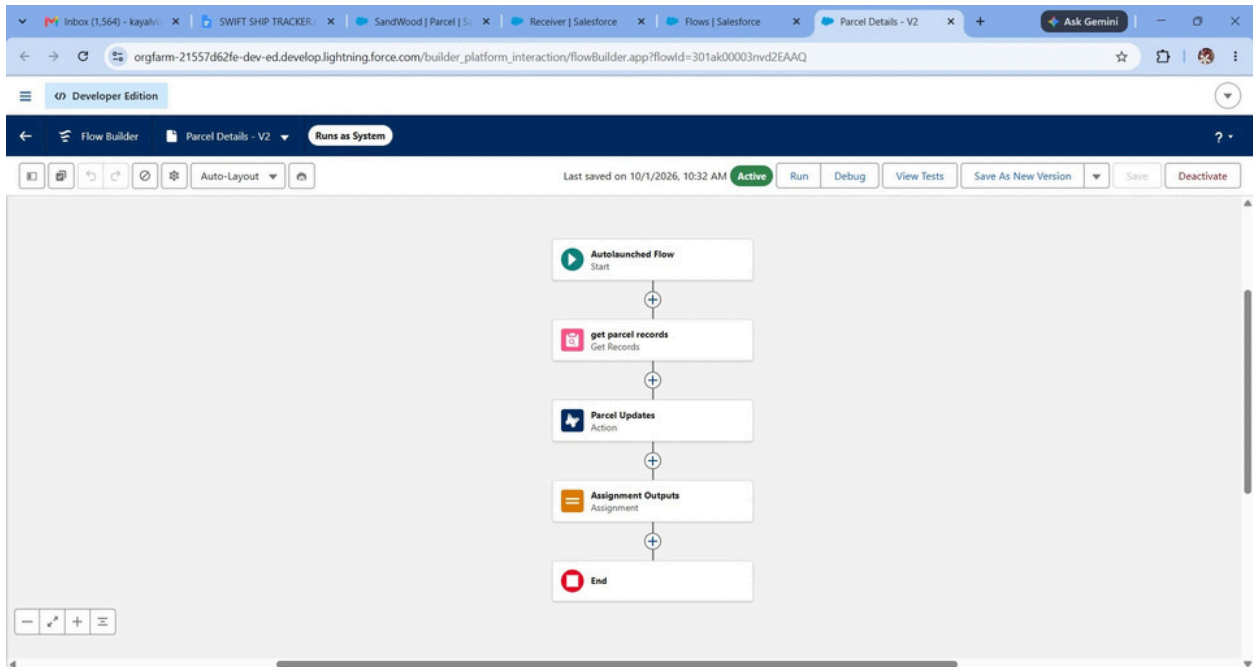
Start → Get Records → Action → Assignment → End

8. Save the Flow.
9. Name the Flow: Parcel Details.
10. Activate the Flow.

Configure Flow User Access

11. Once the Flow is saved, go back to the Flows page.
12. Click the dropdown near the Flow that was just created.
13. Click Edit Access.
14. At the top, look for Flow User Access.
15. Assign the available profiles as required.

16. Click Save.



PHASE 3: UI/UX DEVELOPMENT & CUSTOMIZATION

To enable agents to efficiently manage parcel operations and ensure secure access control using permission sets and AI-driven agent experiences built with AgentForce.

Milestone 1 – Creating PermissionsSet

In the Quick Find box on the left side, type Permission Sets.

1. Click Permission Sets under Users.
2. Click the New button.
3. Enter the following details:
 - Label: Swift Ship
 - API Name: Auto-fills
 - License: Choose Einstein Agent.
4. Click Save.

Configure Object Settings

- 1 Inside the Permission Set page, under App Object Settings, click Object Settings.
 - . Search and click each object one by one.
- 2
- .

3. Click Edit.
4. Enable the following:
 - Read
 - Create
 - Edit
 - Optional: Delete, if you want agents to remove records.
5. Under Field Permissions, ensure at least Visible is checked for all key fields.
6. Click Save after each object.

Configure the following objects:

- Parcel__c

The screenshot shows the Salesforce Setup interface for configuring permissions for the 'Parcels' object. The 'Object Permissions' section is expanded, showing a table of permissions. The 'Field Permissions' section is also expanded, showing a table of field permissions.

Permission Name	Enabled
Read	☒
Create	☒
Edit	☒
Delete	☐
View All Records	☐
Modify All Records	☐
View All Fields	☐

Field Name	Field API Name	Read Access	Edit Access
Created By	CreatedById	☒	☐
Current Location	Current_Location__c	☒	☒
Estimated Delivery	Estimated_Delivery__c	☒	☒
Last Modified By	LastModifiedById	☒	☐
Owner	OwnerId	☒	☐
parcel ID	parcel_ID__c	☒	☐
Parcel Name	Name	☒	☒
promptResponse	promptResponse__c	☒	☒
Sender	Sender__c	☒	☒
Status	Status__c	☒	☒
Weight	Weight__c	☒	☒

- Delivery__c

orgfarm-21557d62fe-dev-ed.develop.my.salesforce-setup.com/lightning/setup/PermSets/page?address=%2F0PSak00000d0hcP%2Fe%3F%3DEntityPermissions%26o%3D01lak00000FHuzd

Developer Edition

Setup Home Object Manager

permission

Users

Permission Set Groups

Permission Sets

Custom Code

Custom Permissions

Didn't find what you're looking for? Try using Global Search.

SETUP Permission Sets

Swift Ship

Find Settings Clone Edit Properties Manage Assignments View Summary

Permission Set Overview Object Settings Deliveries

Deliveries Save Cancel

Tab Settings

Available	Visible
☐	☐

Object Permissions

Permission Name	Enabled
Read	☒
Create	☒
Edit	☒
Delete	☐
View All Records	☐
Modify All Records	☐
View All Fields	☐

Field Permissions

Field Name	Field API Name	Read Access	Edit Access
Created By	CreatedBy	☐	☐
Current Location	Current_Location__s	☒	☒
Delivery Name	Name	☐	☐
Estimated Delivery	Estimated_Delivery__s	☒	☒
Last Modified By	LastModifiedDate	☐	☐
Owner	OwnerId	☐	☐
Parcel	Parcel__s	☒	☒
Sender	Sender__s	☒	☒

- Sender__c

orgfarm-21557d62fe-dev-ed.develop.my.salesforce-setup.com/lightning/setup/PermSets/page?address=%2F0PSak00000d0hcP%2Fe%3F%3DEntityPermissions%26o%3D01lak00000FHv1F

Developer Edition

Setup Home Object Manager

permission

Users

Permission Set Groups

Permission Sets

Custom Code

Custom Permissions

Didn't find what you're looking for? Try using Global Search.

SETUP Permission Sets

Swift Ship

Find Settings Clone Edit Properties Manage Assignments View Summary

Permission Set Overview Object Settings Sender's

Sender's Save Cancel

Tab Settings

Available	Visible
☐	☐

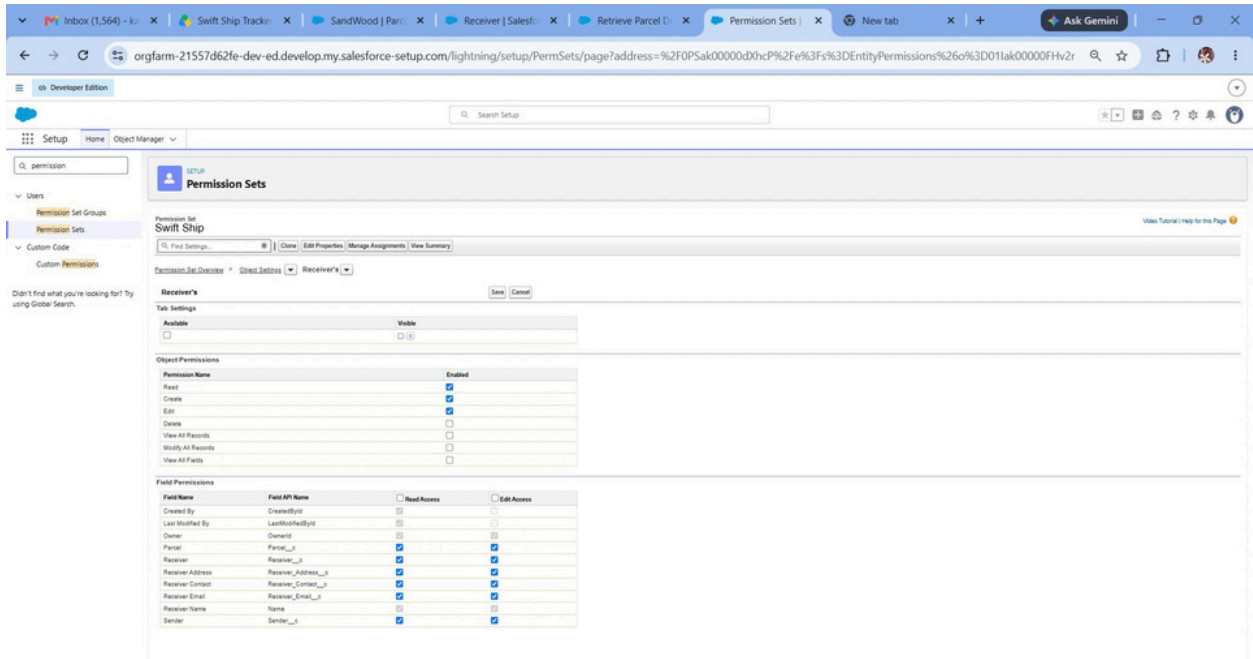
Object Permissions

Permission Name	Enabled
Read	☒
Create	☒
Edit	☒
Delete	☐
View All Records	☐
Modify All Records	☐
View All Fields	☐

Field Permissions

Field Name	Field API Name	Read Access	Edit Access
Created By	CreatedBy	☐	☐
Last Modified By	LastModifiedDate	☐	☐
Owner	OwnerId	☐	☐
Sender Address	Sender_Address__s	☒	☒
Sender Contact	Sender_Contact__s	☒	☒
Sender Email	Sender_Email__s	☒	☒
Sender Name	Name	☐	☐

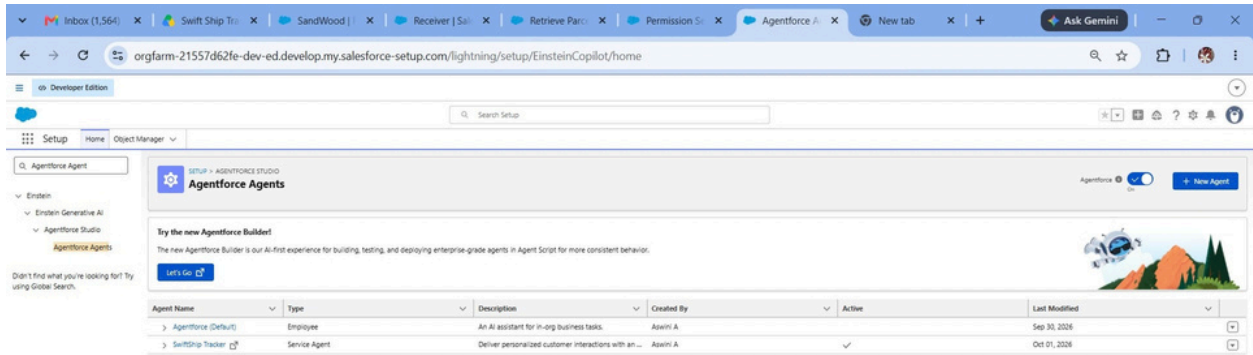
- Receiver__c



Milestone 2 – Assigning Permissions to Your Agent User

In the Quick Find box on the left, type Users.

1. Click Users under the Users section.
2. Open the Agent User Record.
3. From the list of users, find the Einstein Service Agent user to represent your Agent, for example, SwiftShip Agent.
4. Click on the Full Name of that user to open the record.
5. Scroll down to the section named Permission Set Assignments.
6. Click the Edit Assignments button.
7. On the left side under Available Permission Sets, find and select your permission set:
 - SwiftShip
8. Click the Add → button to move it to the right-side list under Enabled Permission Sets.
9. Click Save.



Milestone 3 – Org Setup for AgentForce

Enable Salesforce Default Agent

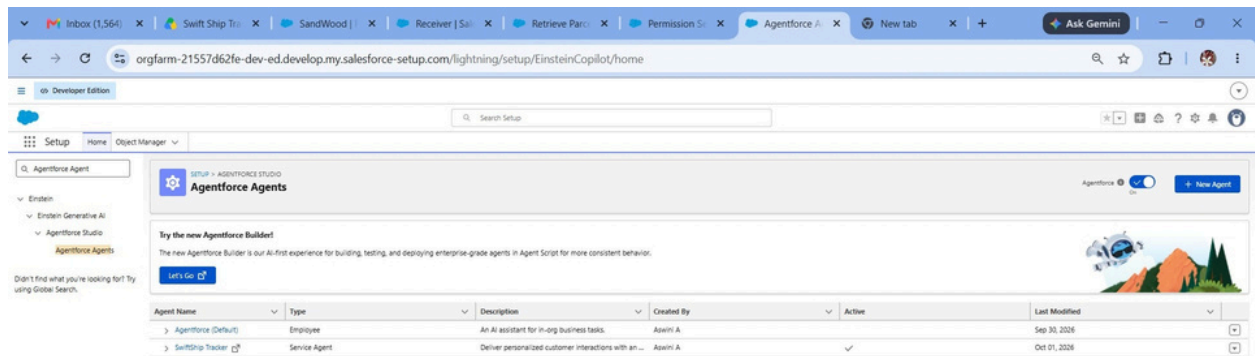
1. From the Quick Find box, search for Salesforce Go.
2. Click on Salesforce Go.
3. Search for Default Agent.
4. Click Get Started and turn it on.
5. Refresh the page.

Enable Einstein Setup

Go to Setup → Quick Find → Einstein Setup.

1. Open Einstein Setup.
2. Click Get Started if prompted.
3. Turn on the following key services:
 - Turn On Einstein
 - Global Languages
 - Deploy Prompt Template
4. Refresh your browser page.
5. Go to Setup → Quick Find → AgentForce Agents.
6. Turn on:
 - Agentforce

- Enable Agentforce Default Agent

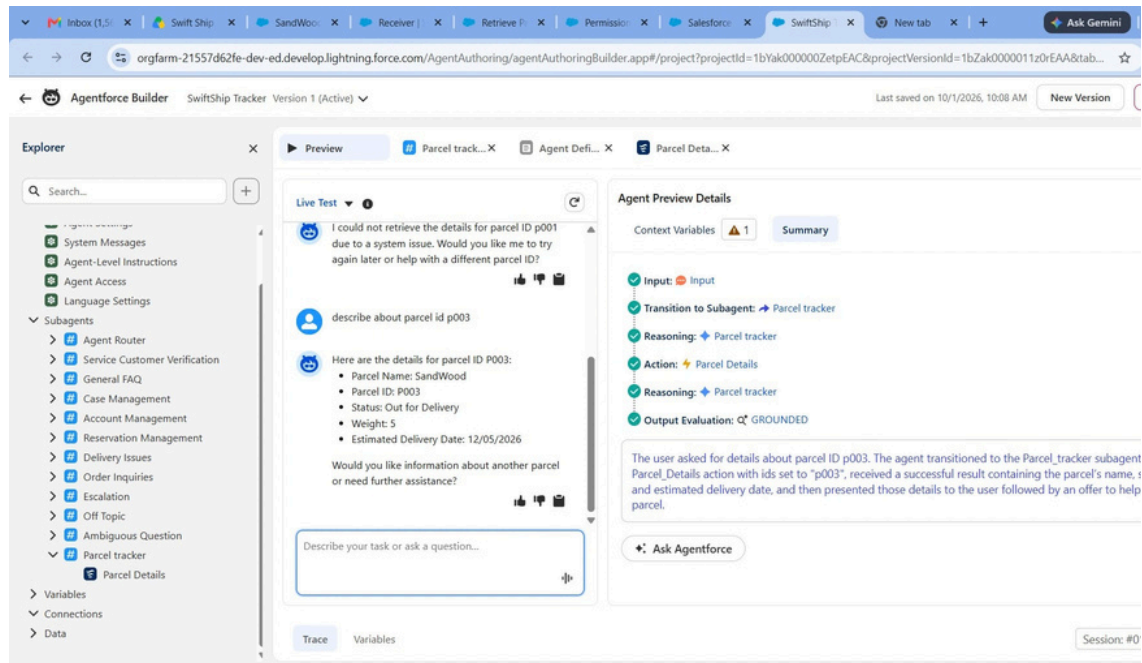


Milestone 4 – Building the Agent

- 1 In Setup, search for Agentforce Agents in Quick Find.
 - . Click Agentforce Agents.
- 2 Click on the Default Agent.
 - . Click Open in Builder.
- 3

Assign Sub-Agents to the Agent

- 4
 - 1 In the Subagents section, click New.
 - . Click New Subagent.
 - 2 Create a new subagent.
 - . Fill in the details as shown in the document.
 - 3
 - .
 - 4
 - .



Adding Actions to Agentforce Assets

While still in Agent Builder:

1. Go to the Topics Actions section.
2. Click New Agent Action.
3. For Action Type, select Flow.
4. Click Next.
5. For Reference Flow, select the recently created Flow.
6. Click Next.
7. Fill in the details for Agent Configurations.

Assign Subagent Action

1. While in Agent Builder, go to Subagent.
2. Click on the Parcel Updates Subagent.
3. Click on Subagent Actions.
4. Select Add from Asset Library.
5. Select the Flow action that was created.
6. Click Finish.
7. Click Settings.
8. Rename your agent to SwiftShip Tracker.
9. Click Save.

10. Click Ignore and Activate.
11. Click Activate.

PHASE 4: DATA MIGRATION, TESTING & SECURITY

To validate the functionality and reliability of the Parcel Tracking Agent created in Agent Builder.

This phase ensures that the agent correctly recognizes topics, triggers the assigned Flow (SwiftShip Tracker), retrieves or updates parcel data, and performs all intended actions without access or permission issues.

The goal is to confirm that the AI-driven agent operates seamlessly within Salesforce and provides accurate, real-time responses to parcel management queries.

Verify Agent

Go to Setup → Agentforce Agents.

1. Open the Agents list and confirm that your agent appears.
2. Agent Name: Parcel Tracking Agent

Validate Assigned Topic

In Agent Builder → SubAgent, confirm that the SubAgent you created, such as Parcel Tracking or SwiftShip Tracker, is listed.

Open the topic and verify:

- It contains a prompt or description related to parcel management.
- The Action/Flow (SwiftShip Tracker) is linked to this topic.
- Click Save and ensure that the topic is Active.

Check the Flow Association

Go to Setup → Flows.

1. Locate the Flow named SwiftShip Tracker.
2. Confirm:
 - The Flow is Activated.

- TheFlow includes screens or actions that handle:

Parcel ID input

Parcel details retrieval

Status update or confirmation message

3. Open the Flow's Advanced Settings.
4. Ensure Run in System Context (Without Sharing) if data access is restricted.

Activate and Test the Agent

1. Return to Setup → Agent Builder → Parcel Tracking Agent.
2. Click Activate, if it is not already activated.
3. Once active, test using Preview / Try it Out mode.
4. Type a query such as:

"Track parcel ID P12345"

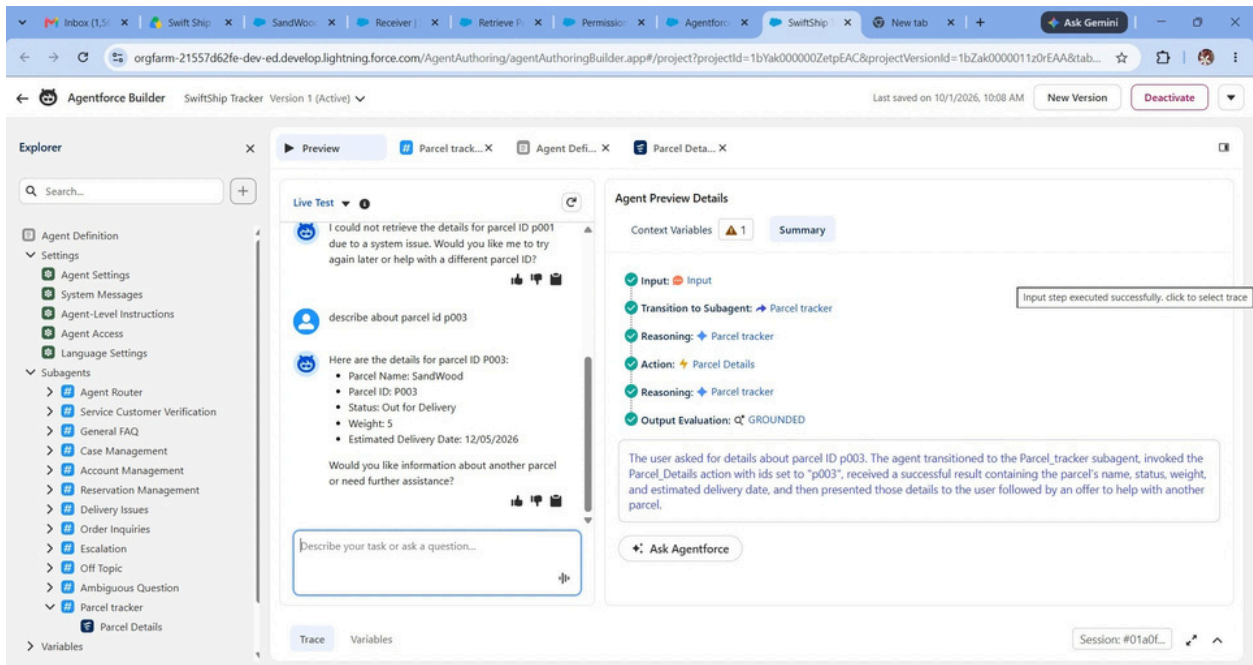
or

"Update delivery status for parcel P67890 to Delivered."

5. Observe the agent's response.

Validate Data in Salesforce

1. After the agent runs the Flow:
 - Go to App Launcher → Parcels (Parcel__c).
2. Search for the Parcel ID used during testing.
3. Verify:
 - The Status or related field is updated.
 - Any Flow-created records, bookings, or updates appear correctly.



PHASE 5: DEPLOYMENT, DOCUMENTATION & MAINTENANCE

In the scope of our SwiftShip Tracker (AgentForce) project implementation:

- The purpose of this phase is to validate that the Parcel Tracking Agent, created using Salesforce Agent Builder, works as intended within the Salesforce Developer Org environment.
- The main goal is to understand how an AI-powered Service Agent can be tested to ensure correct topic recognition, Flow execution, and data retrieval from Salesforce custom objects.
- Testing is conducted within the Developer Org to simulate how the Agent would perform in a live setup, ensuring that permissions, flows, and interactions function correctly before any deployment stage.
- Real-world production testing (UAT or sandbox deployment) is not performed in this project because:
 - Developer Edition orgs are standalone environments and do not support production-linked testing environments.
 - Enterprise-level AgentForce deployment involves sandbox testing, staging environments, and CI/CD automation, which are outside the scope of this academic/prototype implementation.

Maintenance, Monitoring & Troubleshooting

In the SwiftShip Tracker (AgentForce) project, ongoing maintenance ensures that Parcel, Customer, and Delivery data remain accurate, synchronized, and up to date.

Regular monitoring helps detect issues such as misrouted agent queries, failed Flow executions, or performance delays in AgentForce responses.

Troubleshooting activities include resolving configuration errors, updating Flow logic, refining agent topics, and improving prompts to enhance user interactions.

Although this project is implemented within a Salesforce Developer Org for learning purposes, in real-world enterprise deployments, these maintenance and monitoring practices are essential. They ensure that AI-powered service agents remain reliable, scalable, and continuously aligned with evolving customer service operations and business goals.

FUNCTIONAL & PERFORMANCE TESTING

During testing, screenshots were captured from Salesforce to validate vehicle creation, dealer assignment, stock validation triggers, order status updates, batch processing, scheduled reminders, and dashboard analytics.

All features were verified to ensure correct functionality and performance.

ADVANTAGES & DISADVANTAGES

Advantages

- Centralized parcel and customer information in Salesforce.
- Provides better visibility into parcel booking and delivery tracking.
- Automates key logistics processes using Salesforce Flow.
- Agentforce enables conversational customer support and parcel-status assistance.
- Prompt Builder supports AI-powered responses and self-service.
- Reduces manual effort for customers, support teams, and administrators.
- Provides timely notifications and updates during the delivery process.
- Experience Cloud can provide customers with convenient access to tracking information.
- Supports integration with external logistics systems.
- Can scale to support multiple locations and growing parcel operations.

Disadvantages

- Requires accurate and up-to-date parcel and customer data.
- Dependence on Salesforce configuration and automation.
- Agentforce responses depend on properly configured prompts and available data.
- External logistics integrations may increase implementation complexity.
- Flow and automation logic require regular maintenance when business processes change. Users need appropriate Salesforce or Experience Cloud access to use the system.
- Initial Salesforce implementation and configuration can require significant effort.
- Expanding the system to additional locations or integrations may require further customization.

CONCLUSION

The SwiftShip Tracker demonstrates how Salesforce can be used to build a centralized parcel management and tracking solution. The project integrates custom objects, relationships, Flows, Apex, Email Alerts, Experience Cloud, AgentForce AI, Prompt Builder, security controls, and reporting to support parcel booking, delivery tracking, status updates, and customer communication.

The integration of AgentForce AI with Prompt Builder and Salesforce Flow provides a conversational approach to parcel tracking, enabling customers to obtain shipment information without depending entirely on manual support.

The project therefore provides a practical foundation for a scalable and user-friendly parcel management solution.

FUTURE SCOPE

- Extend AgentForce capabilities for more parcel-management queries.
- Enhance real-time delivery tracking.
- Expand customer self-service through Experience Cloud.
- Add more advanced delivery-performance analytics.
- Integrate additional external logistics systems.
- Improve AI prompts and conversational experiences.
- Introduce enterprise-level sandbox and CI/CD deployment.
- Expand automation for additional parcel operations.

The source identifies enterprise deployment, continuous monitoring, prompt refinement, Flow improvement, and AI-agent scalability as areas for future enhancement.

APPENDIX

- A. Custom Objects: Parcel, Delivery, Sender, Receiver
- B. Standard Objects: Account, Contact, Case, Task, EmailMessage, User
- C. Automation: Flows, Approval Processes, Email Alerts
- D. Apex: Apex Classes, Batch Apex
- E. AI: AgentForce AI, Prompt Builder
- F. UI: Lightning App, Lightning Record Pages, Experience Cloud
- G. Security: Profiles, Roles, Permission Sets, Sharing Rules, Field-Level Security
- H. Analytics: Reports & Dashboards
- I. Testing: Flow, AgentForce, data retrieval, security and permission validation